Robust Contactless Palmprint Verification with Deep Metric Learning: Checkpoint 2 — Advanced Model, Baseline Comparison, and Robustness Analysis

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***Abstract*—Checkpoint 1 established the data foundation and an initial ResNet18 + ArcFace embedding model for contactless palmprint verification. Checkpoint 2 takes a technical deep dive: we present a full system architecture, expand the methodology with formal loss and scoring definitions, survey 5–7 related state-of-the-art methods, and benchmark the advanced model against a realistic baseline floor. On the PolyU–IITD dataset (611 subjects, 12,220 images, subject-disjoint splits), our advanced model reaches an Equal Error Rate (EER) of 1.67% on clean test data, a 3.9× reduction over an ImageNet-pretrained ResNet18 baseline (EER 6.50%). A systematic benchmark across 9 corruption types × 5 severity levels exposes critical vulnerabilities: noise (+1,170% EER), motion blur (+1,041%), and brightness shifts (+796%) cause the steepest drops, while contrast and scale changes remain tractable. These targeted weaknesses motivate the mitigation work planned for the final phase.**

I. Refined Problem Statement

As established in Checkpoint 1, this project investigates the robustness of a deep learning–based palmprint verification system to common capture variations. Checkpoint 2 extends that work by (i) formalizing the full system architecture, (ii) articulating the mathematical formalism of the training objective and matcher, (iii) comparing the advanced model to a quantitative baseline floor, and (iv) presenting a full robustness benchmark.

A. Scope Changes Since Checkpoint 1

- **Dropped:** MobileNetV3 backbone comparison—effort was reinvested in a thorough robustness benchmark and a rigorous baseline floor.
- **Added:** An ImageNet-pretrained-only ResNet18 *baseline floor* (no fine-tuning) scored on the identical test pair set, providing a concrete non-trivial reference point.
- **Completed:** Full robustness benchmark across 9 corruption types × 5 severity levels (45 conditions), exceeding the originally planned 4–6 types.
- **Deferred:** Mitigation (robust augmentation training or quality gating) is deferred to the final phase.

II. System Architecture

Fig. 1 diagrams the end-to-end verification pipeline, with the five labeled stages required of a biometric system: input interface, preprocessing pipeline, feature extraction and template generation, matcher, and database store.

a) Input interface: Raw input is the PolyU–IITD contactless palmprint sensor output, pre-cropped to 128×128 grayscale ROI.

b) Preprocessing pipeline: Grayscale frames are expanded to 3-channel RGB for ImageNet-pretrained backbones and normalized with ImageNet channel statistics. During training, augmentation (random crop, horizontal flip, $\pm 15^\circ$ rotation, color jitter, Gaussian blur, random erasing) reduces overfitting.

c) Feature extraction & template generation: A ResNet18 backbone extracts 512-D features; a two-layer MLP projector with batch normalization and ReLU reduces this to a 256-D embedding; L2 normalization places the template on the unit hypersphere, making cosine similarity the natural matcher score.

d) Matcher/Classifier: At inference, two templates $f(x_p)$, $f(x_q)$ are compared by cosine similarity; a decision threshold τ (set at the EER operating point, $\tau \approx 0.30$) decides accept or reject. At training time, an ArcFace classification head [2] with scale $s = 30$, margin $m = 0.3$, imposes an angular margin between classes to sharpen inter-identity separation.

e) Database store: Enrolled templates are indexed by (subject, hand) class label. The store supports 1:1 verification (one gallery template per claim) and 1:N identification (rank all gallery templates by similarity).

III. Data Engineering

A. Dataset Recap

We use the PolyU–IITD Contactless Palmprint Database v3.0 [1] with 611 subjects (12,220 images), subject-disjoint splits (427/61/123 train/val/test), and each (subject, hand) pair as a unique class. Four empty placeholder subjects (H_ID608–611) identified in Checkpoint 1 are excluded.

B. Verification Pair Generation

For evaluation, we generate verification pairs from the test set (123 subjects, 246 classes, 2,460 images): 5,000 genuine pairs (same class) and 5,000 impostor pairs (different classes),

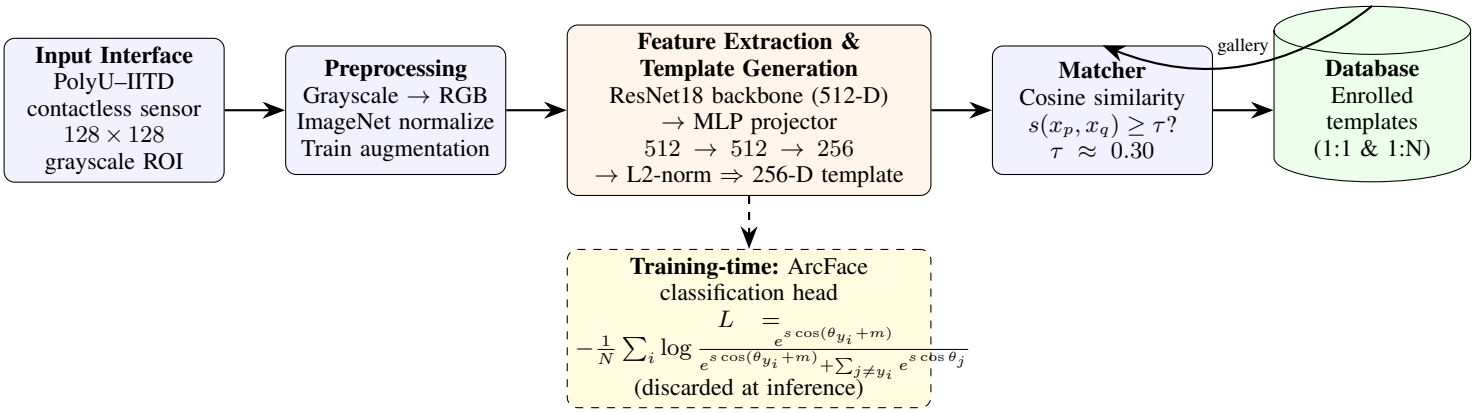

Fig. 1. System architecture for palmprint verification. Input images are normalized and passed through a ResNet18 backbone and projection head that produce L2-normalized 256-D templates. At enrollment, templates are stored in the database; at verification, a query template is compared against a gallery entry via cosine similarity and thresholded. The ArcFace classification head (dashed) is used only during training to structure the embedding space and is discarded at inference.

seeded at 42 for reproducibility. Each pair is scored by cosine similarity between L2-normalized embeddings. The same pair set is reused across all corruption conditions and across the baseline/advanced comparison for consistency.

IV. Related Work

Palmprint recognition research spans three lines of work that directly inform this project: handcrafted palmprint coding, deep palmprint recognition, and the broader deep metric learning / robustness literature.

A. Handcrafted Palmprint Coding

Early palmprint verification is dominated by filter-response coding. Zhang et al. introduced *PalmCode* [6], encoding Gabor-filter phases into binary templates matched by Hamming distance; Kong and Zhang's *Competitive Code* [7] instead records the *direction* of the dominant Gabor response. Guo et al.'s *BOCV* [8] extends this with multi-orientation binary coding. These methods are fast and interpretable but brittle to pose, illumination, and sensor changes because the features are hand-designed. Our approach replaces the handcrafted filterbank with a learned ResNet18 feature extractor, trading interpretability for robustness.

B. Deep Palmprint Recognition

Genovese et al.'s *PalmNet* [9] pretrains Gabor-inspired convolutional filters and selects them via PCA, bridging handcrafted and deep approaches. Matkowski et al. [10] focus on contactless palmprint ROI extraction and cross-domain matching using CNNs trained end-to-end. Both use classification or contrastive losses without explicit angular margin. Our system adopts a cleaner two-stage architecture—backbone + projection head on the unit hypersphere—and supervises it with ArcFace, which is empirically stronger for identity-style tasks.

C. Deep Metric Learning and Robustness

Schroff et al.'s *FaceNet* [5] popularized triplet-loss embedding learning for face verification, and Deng et al.'s *ArcFace* [2] replaced triplets with a classification-based additive angular margin that is simpler to optimize and yields better separation; we adopt ArcFace directly. Hendrycks and Dietterich's *ImageNet-C* [4] formalized the practice of benchmarking corruption robustness at multiple severity levels across noise, blur, weather, and digital artifacts; our 9×5 robustness benchmark adapts their protocol to the palmprint domain. Compared to these works, our contribution is the *systematic* measurement of how a SOTA face-verification recipe (ArcFace) degrades under palmprint-relevant corruptions, and the identification of the specific failure modes that a production system would need to mitigate.

V. Advanced Methodology

A. Architecture Choice

The advanced model is a two-stage network with ∼12M parameters. The backbone is an ImageNet-pretrained ResNet18 [3] (Conv1 through Conv5 residual stages, global average pooling) with its final fully-connected classification layer replaced by an identity mapping, yielding a 512-D feature. A projection head

$$512 \xrightarrow{\text{Linear}} 512 \xrightarrow{\text{BN, ReLU, Dropout}(p=0.5)} 256 \xrightarrow{\text{BN}} \text{L2-norm}$$

emits a 256-D template $f(x) \in \mathbb{S}^{255}$ on the unit hypersphere, making cosine similarity the natural matcher score. During training, an ArcFace head [2] with scale $s = 30$ and angular margin $m = 0.3$ is attached; at inference it is discarded.

B. Training Strategy

The model is optimized with AdamW (learning rate 3×10^{-4}, weight decay 10^{-3}), batch size 32, with label smoothing 0.1 and a 5-epoch warmup followed by cosine decay to 10^{-6} over 50 epochs. Early stopping (patience 10, min-delta 10^{-4} on

validation EER) selected the checkpoint at epoch 32 (validation EER 1.13%); training halted at epoch 42. Augmentation uses random crop, horizontal flip, $\pm 15^\circ$ rotation, random affine translation/scale/shear, color jitter, random grayscale, Gaussian blur, and random erasing (p=0.2). The seed is fixed at 42 across splits, pair generation, and initialization.

C. Mathematical Formalism

a) ArcFace loss.: Let $f(x_i) \in \mathbb{R}^d$ denote the L2-normalized embedding of input x_i with ground-truth class y_i, and let $W \in \mathbb{R}^{C\times d}$ be the row-normalized classifier weights, so $W_j^\top f(x_i) = \cos\theta_{i,j}$. With scale s and additive angular margin m, the per-sample ArcFace loss is

$$L_{\text{AF}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{s\cos(\theta_{i,y_i}+m)}}{e^{s\cos(\theta_{i,y_i}+m)} + \sum_{j\neq y_i} e^{s\cos\theta_{i,j}}}. \quad (1)$$

Intuitively, the margin m enforces a wider gap between the ground-truth class and all others *on the hypersphere*, which directly translates to better cosine-similarity separation at test time.

b) Verification rule.: At inference, a probe x_p and gallery template x_q are accepted as the same identity iff

$$s(x_p, x_q) = \frac{f(x_p)^\top f(x_q)}{\|f(x_p)\|\,\|f(x_q)\|} \geq \tau, \quad (2)$$

where τ is tuned on a held-out set. Because f is L2-normalized, $s(x_p, x_q) = f(x_p)^\top f(x_q) \in [-1, 1]$ and (2) reduces to a single inner product.

c) Evaluation metrics.: Given scored genuine/impostor pairs, the False Acceptance Rate (FAR) and False Rejection Rate (FRR) at threshold τ are

$$\text{FAR}(\tau) = \Pr[s \geq \tau \mid \text{impostor}], \quad \text{FRR}(\tau) = \Pr[s < \tau \mid \text{genuine}] \quad (3)$$

and the Equal Error Rate is $\text{EER} = \text{FAR}(\tau^\star) = \text{FRR}(\tau^\star)$ at the crossing $\tau^\star$.

VI. Exploratory Data Analysis

Fig. 2 shows the genuine/impostor cosine similarity distributions on clean test data. Genuine pairs concentrate around 0.759 ± 0.160, impostor pairs around 0.033 ± 0.105; the narrow overlap near 0.2–0.4 determines the EER. Fig. 3 projects 256-D embeddings for 20 test identities via t-SNE, showing tight, well-separated clusters consistent with the low EER. Class balance is maintained by design (10 images per hand per subject).

VII. Preliminary Results and Comparison

A. Baseline Floor vs. Advanced Model

To quantify the contribution of deep metric learning, we compare the advanced model against a realistic floor: an ImageNet-pretrained ResNet18 *without* palmprint-specific fine-tuning. Features (512-D, L2-normalized) are scored by cosine similarity on the identical test pair set (seed 42). Both systems are evaluated on 5,000 genuine and 5,000 impostor pairs.

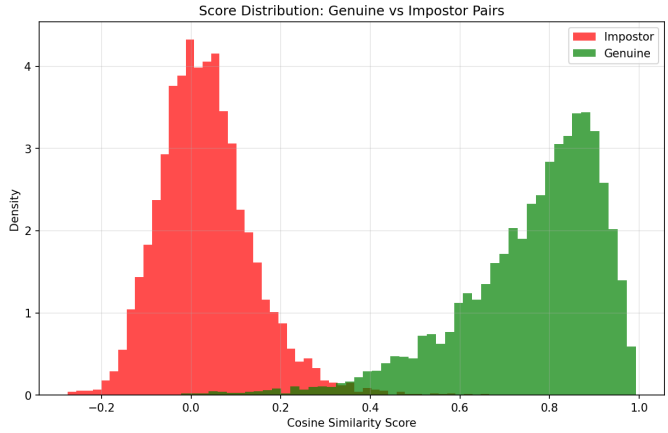

Fig. 2. Genuine (green) vs. impostor (red) cosine similarity distributions on the test set. Clear bimodal separation enables reliable verification at $\tau \approx 0.30$.

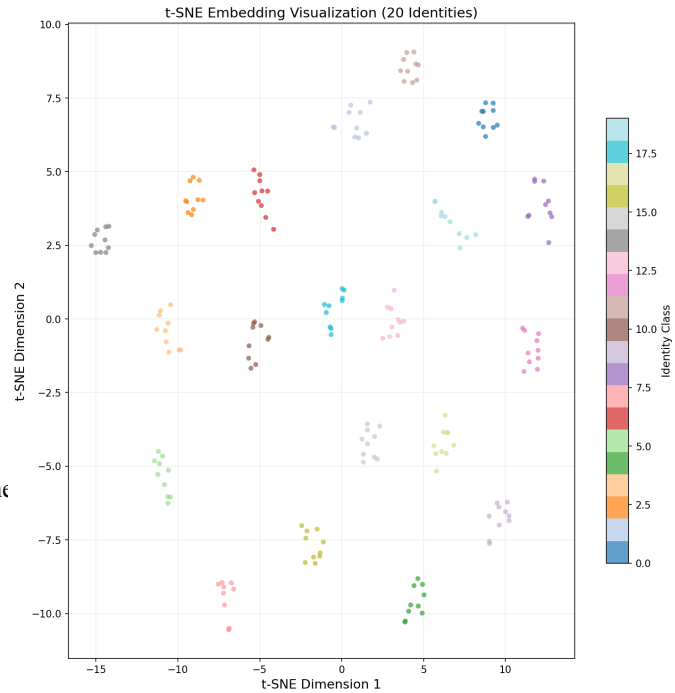

Fig. 3. t-SNE visualization of 256-D advanced embeddings for 20 test identities. Tight, well-separated clusters confirm discriminative embedding learning.

The advanced model cuts EER by 3.9× (6.50% → 1.67%) and FAR at 1% FRR by 7.0× (28.74% → 4.12%). The impostor mean drops from 0.677 to 0.033, showing that the ArcFace objective successfully pushes apart the angular positions of different identities on the hypersphere; the pretrained features, by contrast, collapse all palmprint images into a narrow cone of ImageNet-inherited features.

B. ROC and Threshold Analysis

Fig. 4 shows the ROC curve for the advanced model (AUC $>$ 0.99). The EER operating point corresponds to $\tau \approx 0.30$. The high FAR @ 0.1% FRR (48.88%) reveals overlapping

TABLE I
BASELINE FLOOR VS. ADVANCED MODEL ON CLEAN TEST DATA.

Metric	Pretrained ResNet18 (baseline floor)	ResNet18 + ArcFace (advanced)
EER	6.50%	**1.67%**
FAR @ 1% FRR	28.74%	**4.12%**
FAR @ 0.1% FRR	47.52%	**48.88%**†
Genuine μ	0.914	0.759
Impostor μ	0.677	0.033

†Floor nearly matches at this strict operating point because both suffer from overlapping tails; EER and FAR@1%FRR are the primary metrics.

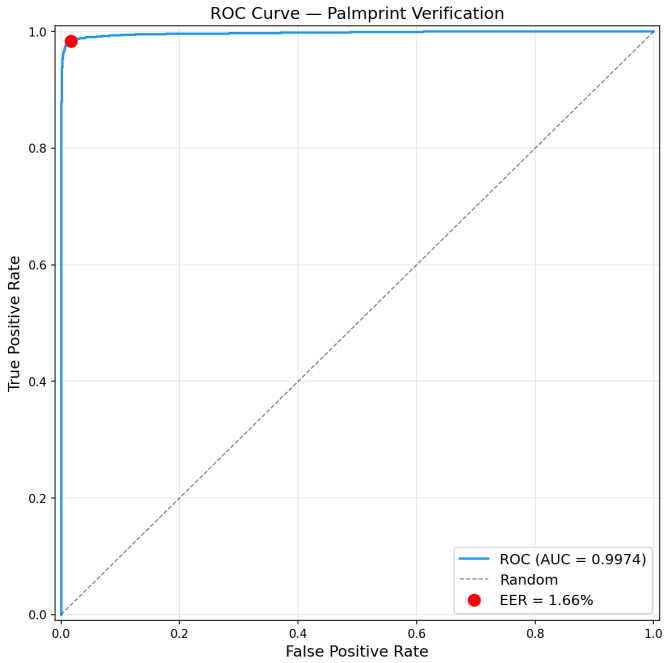

Fig. 4. ROC curve of the advanced model with EER operating point (red dot). AUC > 0.99.

distribution tails: at very strict operating points required for high-security deployments, the system cannot separate all pairs. This likely stems from limited intra-class variability in training ($\sim$10 images per class), which constrains the model's coverage of within-identity variation.

C. Robustness Benchmark

We re-evaluate the advanced model under 9 corruption types at 5 severity levels (45 conditions); corrupted images are generated on the fly from the clean test set and the same 10,000 pairs are re-scored. Table II ranks corruption types by worst-case degradation relative to the clean EER of 1.67% (1.72% measured on the corruption pipeline's clean baseline).

D. Error Analysis

The three worst corruptions—noise (+1,170%), motion blur (+1,041%), and brightness (+796%)—each drive EER above 15% at severity 5. The failure mechanism is consistent: these corruptions destroy or mask the fine-grained texture cues (principal lines, wrinkles) that dominate the learned representation.

TABLE II
ROBUSTNESS BENCHMARK ON THE ADVANCED MODEL: EER DEGRADATION BY CORRUPTION TYPE.

Corruption	Mean EER	Worst EER	Rel. Degrad.
Motion Blur	10.74%	19.67%	+1,041%
Noise	9.00%	21.90%	+1,170%
Brightness	5.45%	15.45%	+796%
Occlusion	5.29%	7.80%	+352%
Rotation	4.52%	9.15%	+430%
JPEG	4.19%	11.12%	+545%
Gauss. Blur	2.83%	4.45%	+158%
Scale	2.75%	3.05%	+77%
Contrast	1.85%	2.28%	+32%

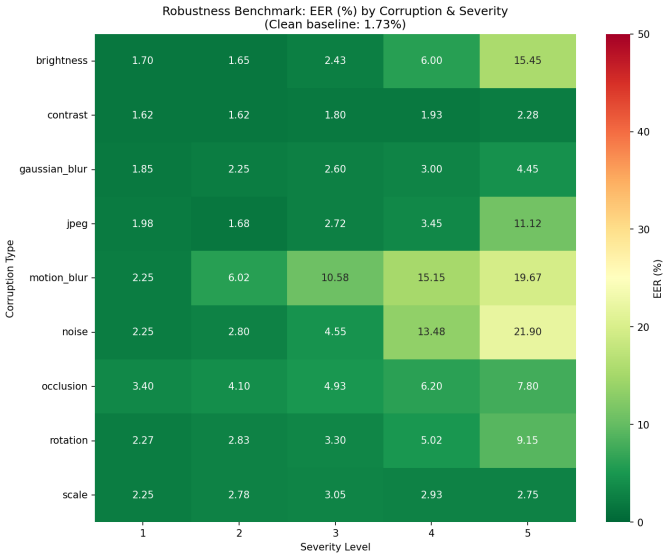

Fig. 5. Robustness EER heatmap: 9 corruption types $\times$ 5 severity levels. Darker cells indicate higher EER.

Qualitatively, failure cases concentrate in three regimes: (i) *high-noise probes* where the impostor similarity spikes because Gaussian noise fills the high-frequency band the network relies on, (ii) *motion-blurred genuine pairs* where intra-class similarity collapses toward the impostor mode as line structure is smeared, and (iii) *over-exposed samples* where saturated pixels erase texture entirely. By contrast, contrast changes (+32%) and scale changes (+77%) are nearly absorbed—contrast by batch normalization and ImageNet normalization, scale by the training-time random crop and affine augmentation. These asymmetries are the clearest signal for the mitigation phase: a targeted mix of noise/blur/brightness augmentation or a dedicated quality gate should address the dominant failure modes.

VIII. CODE REPOSITORY

Source code is at https://github.com/onggiahuy97/cs228-palmprint.

Repository layout:

- `src/` – training, evaluation, corruption, baseline, visualization

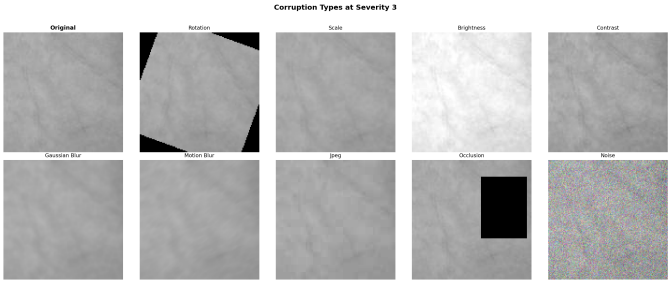

Fig. 6. Original palmprint and 9 corruption types at severity 3. Noise, motion blur, and brightness cause the steepest EER growth at severity 5.

- `splits/` – subject-disjoint split definitions (JSON + TXT)
- `figures/` – ROC, t-SNE, training curves, corruption samples
- `checkpoints/baseline_v2/` – advanced model weights, per-epoch snapshots, clean-test logs, `robustness/` CSVs and plots
- `checkpoints/baseline_floor/` – pretrained-only floor `metrics.json`
- `report/` – this document

Key scripts:

- `src/train.py` – ArcFace training loop with early stopping
- `src/evaluate.py` – verification pair generation + EER/FAR/FRR
- `src/robustness_eval.py` – 45-condition corruption benchmark
- `src/baseline_floor.py` – pretrained-only ResNet18 baseline
- `src/corruptions.py` – 9 corruption functions $\times$ 5 severities
- `src/model.py` – ResNet18 embedder + ArcFace head

References

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[10] W. M. Matkowski, T. Chai, and A. W. K. Kong, "Palmprint recognition in uncontrolled and uncooperative environment," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 1601–1615, 2020.

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